Perspectives on Pain in the Low Back and Neck: Global Burden, Epidemiology, and Management

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Introduction

Low back pain (LBP) is a major public health problem worldwide. All age groups are affected, including children and adolescents,[1] with 1%-2% of adults in the United States being disabled as a result.[2] Back and neck pain (NP) are two of the most common reasons for visits to primary care physicians and chiropractors in the United States and cause considerable disability and financial burden.[3] Although people don't die from low back or neck pain, the morbidity toll is enormous from both personal and societal perspectives. Both are reported by more patients and have a higher impact in the workforce, as well as financially, than any other musculoskeletal disorder and most other clinical conditions.[4] In 2005 only heart disease and stroke had substantially higher medical expenditures than spine disorders in the United States.[5] The large and rising spine-related health-care expenditures over the last decade do not seem to be associated with improved self-assessed health status, or improvement in functional ability, work limitations or social functioning.[5] In this article, we take a step back to look at this toll, and look ahead to see where we, as individuals and as a society, should go in order to reduce the burden of back and neck pain.

Global Burden of Disease — Low Back and Neck Pain

In studies that measure the Burden of Disease (BoD), diseases are ranked according to how much death and disability they cause.[6] The BoD estimates are useful to governments and organizations when planning for health priorities, usage of resources, and assessing costs or benefits of interventions in the public health sector.[6] Disability adjusted life years (DALYs) is an overall summary measure of population health that the global BoD uses, which combines years of life lost due to premature mortality (YLLs) and years lived with disability (YLDs).[7]

LBP is presently the leading cause of disability in the world.[7] Case definitions of LBP are quite variable, thus estimation of the global burden is not as easy as it would seem.[7] According to the Global Burden of Disease (GBD) 2010, LBP results in more disability than any other condition worldwide (10.7% of total YLDs; ranks 6th in terms of overall disease burden [83 million DALYs]); and is the leading cause of disability in both developed and developing countries.[8] These estimates are up significantly from the GBD 2000-2004 estimates (eg, LBP was ranked 105 out of 136 conditions in terms of YLDs); however, as LBP was defined differently, the GBD estimates cannot be used to assess trends over time.[7] Given the above estimates, however, LBP is a condition that demands our attention in research, public health, and patient health-care.

NP is also one of the major causes of disability globally. The 2010 GBD estimates reflect the fact that both NP prevalence and burden is high around the world. As case definitions of NP are also quite variable, the NP global burden is not easy to estimate either; however, it is based on an extensive series of systematic reviews capturing a large number of studies. Among 291 other health conditions, NP ranked 4th in terms of disability (in YLDs) and 21st in overall burden of disease (in DALYs).[7]

Epidemiology of Low Back Pain

Major indicators of disease occurrence used in epidemiologic studies are incidence (number of new cases in a given period of time) and prevalence (number of individuals having the disease at a given point or period in time).[9] Although LBP is often seen as a condition with individual events characterized by episode duration (eg, acute: less than 6 weeks; sub-acute: 6-12 weeks; chronic: more than 12 weeks)[10] epidemiologic studies are not suggestive of a model of back pain as a series of separate unrelated events but rather as a long-term condition.[11] Currently there are discussions of models that follow the life-course epidemiology approach, exploring long-term processes (ie, biological, behavioural, psychosocial, etc.) that associate adult health conditions with exposures in gestation, childhood and adolescence, earlier adulthood and even across generations.[11] It is quite clear that back pain for example is common among children and especially in adolescents, suggesting that episodes of pain in adulthood might have resulted from earlier in life exposures.[11]

Prevalence and Incidence

Most people will experience LBP at some point in their lifetime, with two-thirds having a recurrence and one third having periods

of disability.[4] One-year recurrence estimates range from 24% to 80%.[12] As longitudinal studies are much more expensive than cross-sectional ones, there are multiple studies reporting on LBP prevalence but fewer reporting on incidence. The 1-year first-ever LBP incidence has been estimated between 6.3% and 15.4%, while the 1-year (overall) incidence ranges from 1.5% to 36% (ie, first-ever or recurrent LBP incidence).[12] However, we need to note that these studies only take into account the first episode in the year and do not consider repeat episodes, resulting likely in underestimation of LBP episode incidence in that time period.

When prevalence is estimated, we are limited in making between-population comparisons, as there is methodological variability across studies due to different case definitions, the recall period, the validation of the instruments used, or the representativeness of the sample, among other issues. Estimates of LBP point prevalence in general population samples range from 1.0% to 58.1% (mean: 18.1%; median: 15%) and 1-year prevalence from 0.8% to 82.5 % (mean 38.1%; median 37.4%).[12] Studies that specify a minimum episode duration (eg, 1 day) have lower prevalence estimates than ones without definitions of episode duration. Heterogeneity in case definitions are known to have large effects on LBP prevalence estimates.[12] For example, descriptions such as "back pain," "low back pain," and "pain on the posterior aspect of the body from lower margin of 12th rib to lower gluteal folds" result in different prevalence estimates. In other systematic reviews, point prevalence ranges from 12% to 33% and 1-year prevalence from 22% to 65%.[13] Prevalence estimates are higher in surveys of self-reported LBP than when medical care data are used (eg, 28%-40% in surveys vs 12%-15% with medical data).[2] Such variable prevalence estimates and the uncertainty with which they have been estimated preclude accurate estimates of prevalence change over time.[2]

Recurrence, Duration, and Remission

LBP can be described as a long-term, recurrent condition that follows many different trajectories rather than as an acute, sub-acute or chronic condition.[11,14] Cases in which LBP never recurs are rare; many patients, suffering between episodes, change what they do to manage recurrences. LBP episode duration estimates range from a median of 42 days from the start of the episode[15] to 128.5 pain days for LBP lasting between 3 to 6 months.[16] Remission at 1-year has been estimated at 54% and 90%.[12]

Risk and Prognostic Factors

There are several factors that influence the onset and course of LBP. Some of these factors are modifiable, while others are not. A previous episode of back pain is the primary risk factor for a new LBP episode.[11] As reviewed by investigators, age is a risk factor, with prevalence increasing up to 60 or 65 years of age, with reports pointing out that prevalence keeps increasing with increasing age for more severe and disabling back pain.[12] The third decade of life has been noted as having the highest LBP incidence.[12] Although findings are not consistent, some studies have found that LBP prevalence might be higher among older women compared to older men, while others have estimated that women are more likely to take time off work, use LBP-related health care, or develop chronic LBP.[12] Low educational and social status have been associated with higher prevalence and incidence of LBP, as reviewed by authors.[12]

Obesity has been associated with an increased risk of LBP, which might be stronger in women.[17] Psychosocial factors, such as stress, anxiety or depression, have been associated with not only the occurrence but also the transition to chronic LBP, although the direction of the association often is not clear. Workplace psychosocial factors (eg, job dissatisfaction, stress, monotonous tasks, or lack of social support and poor work relations) have been associated with occurrence or transition to chronicity. Although data are limited, physical work demands such as twisting, bending, whole-body vibration or manual handling, are likely LBP risk factors.[18]

Personal and Societal Impacts

LBP is the leading cause of activity limitation and workplace absence in most parts of the world. The consequences of LBP are vast and affect the individual, family, health-care systems, industry, and the economy. This can be attributed to restrictions in physical capabilities, participation, work related and financial burden, use of health-care resources, etc. Such impacts differ depending on access to health care, socio-economic status, and occupation distributions in the community.[12] Direct back pain health-care expenditures in the United States were $90.7 billion in 1998.[19] The burden is even greater when indirect costs such as productivity losses, indemnity pay, litigation, retraining and other administrative costs are considered.[12]

Epidemiology of Neck Pain

NP is a common condition in many regions of the world and is increasing in the general population, as well as, in specific occupations.[9] As with LBP, most people will have NP at some point in time, and also like LBP, case definitions are highly variable. One study notes 300 different definitions used in epidemiologic studies, mainly pertaining to the anatomical region and the recall period.[20]

Prevalence and Incidence

NP is typically first experienced during childhood or adolescence, running an episodic course over time.[9] As with LBP, cross-sectional studies are much more common than longitudinal studies. In a systematic review, the 1-year NP incidence ranged from 10.4% to 21.3%.[9] Studies have shown that NP incidence varies by occupation, with office and computer workers, health-care workers, and transit operators having a high incidence of neck disorders.[21]

Due to considerable methodological heterogeneity in epidemiologic studies it is difficult to compare prevalence estimates between studies in different populations. Researchers have also pointed out that "activity-limiting" NP has a lower prevalence than "any" NP.[9] In a recent systematic review, the NP point prevalence in the general population ranged between 0.4% and

41.5% (mean: 14.4%) while 1-year prevalence ranged between 4.8% and 79.5% (mean: 25.8%), with differences largely resulting from heterogeneity of methods and case definitions.[9]

Remission, Duration, and Recurrence

Many epidemiologic studies define as NP remission, the transition to an asymptomatic state, regardless of future NP episodes. However, it is quite challenging to estimate with accuracy the time to remission. In a systematic review, remission at 1 year was estimated between 33% and 65%[9] and 50%-85% of persons with neck pain in general population samples have reported neck pain 1 to 5 years later.[22] However, a better description of the course of neck pain is needed.

Risk and Prognostic Factors

There are several personal and environmental factors associated with increased risk of NP or its course. Age and NP onset are associated, and several studies have estimated a lower NP incidence in men.[9] Poor self-assessed health, as well as psychological status, previous episodes of NP or LBP, occupation, workplace factors (eg, job dissatisfaction, sedentary work positions, and poor physical work environments), ethnicity, or smoking may be associated with NP onset.[9] Several factors might also be prognostic. It is not clear whether gender is one of those factors, with a possibility for men to be more likely to have a remission over a 1 or 5 year-period. Younger persons are generally more likely to remit, and a previous neck injury, poor self-rated health, poor psychological health, getting angry or frustrated, worrying, or high pain intensity may also be associated with poor NP outcomes.[22]

Personal and Societal Impacts

As noted for LBP, the consequences of NP and related disability are substantial and affect individuals, families, health-care systems, industry, and economies. Patients might have limitations in daily activities such as driving, turning their head, working on the computer, and participating in work, family, community, and sporting activities. Consequences of NP vary depending on several factors related to onset and prognosis, including socioeconomic status, access to health care, and occupations in the community, with low-income countries generally affected more than others.[9] Economic impact is also substantial, especially when both direct and indirect costs are considered, ie, health-care costs, work absenteeism, insurance.

Management

The numerous patients with spine disorders together with our generally poor understanding of etiology has resulted in an overwhelming variety of potential treatments.[2] It is recommended that each patient with LBP or NP go through a screening protocol during their health-care provider visit to rule out less common cases of serious spinal pathology and neurologic conditions, assess patients at higher risk for spinal disorders, and guide any further testing or imaging. The vast majority of cases do not present with serious underlying pathology or other red flags. Multiple therapeutic interventions are available with mostly small effects in the short-term and uncertain effects in the long run. Many interventions are not evidence-based and have serious potential side effects and high complication rates (including death).[2] Spine-related health-care use rates are on the rise for reasons not entirely clear, but probably because of a combination of factors including increasing prevalence of more chronic pain, changing beliefs about pain and pain management, increased use of high-cost interventions, inappropriate and overuse of diagnostic imaging, and rising costs of pharmaceuticals and surgical devices, among other factors.[2] A very important component of patient care is the assessment of treatment outcomes. For instance, there are six main domains relevant to the assessment of LBP: pain symptoms, function, well-being, work, disability, and satisfaction with care,[23] and instruments have been developed to assess these domains. Very important also is the concept of "minimal clinically important change" (MCIC), which represents the smallest individual change score important for the patient. For many instruments measuring pain and disability, a 30% change in a patient's score has been suggested as a satisfactory MCIC for improvement.[24] In addition, the Initiative on Methods, Measurement, and Pain Assessment in Clinical Trials (IMMPACT) group suggests the use of an outcome measure that is a composite of at least two patient outcome measures including pain, function, emotional well-being, and global assessment of improvement.[2]

Management of Low Back Pain

Non-surgical Care: Treatments, Outcomes, and Controversies

There are several non-surgical interventions used for LBP. For acute LBP, for example, interventions include brief education, reassurance, advice to stay active, acetaminophen, non-steroidal anti-inflammatory drugs (NSAIDs), spinal manipulation and other manual therapies, and physical modalities such as heat and ultrasound, among many others.[1] For chronic LBP, recommendations include education, advice to stay active, NSAIDs, exercise, manual therapies, and self-management activities, etc.[1] However, non-surgical interventions have not been without controversies. For example, although being physically active is supported by strong evidence for reducing disability, data are conflicting for the clinical and cost effectiveness of physical therapy in chronic back pain.[2] The use of opioids for chronic LBP is also controversial because of concerns about the potential for abuse, questionable effectiveness, and high mental illness co-morbidity with chronic LBP.[2] People using opioids for back pain are more likely to have underlying anxiety, depression and other psychiatric conditions that impair prognosis.[2]

Surgical Care: Treatments, Outcomes, and Controversies

Interventional spine procedures range from percutaneous injections (eg, epidural steroid injections [ESIs], facet/zygapophysial joint procedures, intradiscal procedures, etc.) to surgery (eg, discectomy, spinal canal decompression, spinal fusion, etc.).[2] Both the techniques used and the usage rates of several

procedures have increased dramatically over the past decade,[2] however supporting evidence for many of these interventions is limited. If for example we take a look at ESIs, their increase in use might be more associated with economic than clinical factors (though short-term pain relief in specific cases can offer substantial clinical benefits).[2]

The frequency of back surgery in the United States is the highest in the world[25] and the rate continues to increase.[26] Spinal fusions increased by 40% between 1998 and 2004 and the rate of complex fusion procedures for lumbar spinal stenosis increased 15-fold between 2002 and 2007.[2] Such increases in spinal fusion cannot easily be explained, but may be related to availability of technologies with inadequate data on their superiority over established techniques with known risk-benefit profiles.[2] When LBP is not accompanied by serious neurologic deficits or for nonradicular persistent LBP, the risks of surgery may outweigh the benefits. However, for radiculopathy with herniated lumbar disc or for spinal stenosis with symptomatology, surgery is indicated in the presence of serious or progressive neurologic deficits.[27] In a recent systematic review on surgery for LBP,[27] fusion was not more effective than intensive rehabilitation for nonradicular back pain, although compared to standard nonsurgical care it was associated with small to moderate benefits. On the other hand, for radiculopathy with herniated lumbar disc and spinal stenosis with symptomatology, surgery was associated with short-term benefits when compared to nonsurgical therapy, although in the long-run such benefits generally declined over time.

Two systematic reviews[28-30] commissioned by the Yale University Open Data Access (YODA) Project used patient level meta-analyses of RCT data obtained by the manufacturer to study the effectiveness and harms of recombinant human bone morphogenetic protein-2 (rhBMP-2) compared to autologous iliac crest bone grafting in spine fusion. Both reviews concluded that there was no clinically important difference in inducing spinal fusion by either the rhBMP-2 or autologous iliac crest bone grafting, and both had similar complications when used in anterior lumbar interbody fusion or posterolateral fusion as summarized by Resnick, et al.[31] Compared to autograft in anterior cervical surgery, however, rhBMP-2 had more complications and ectopic bone formation in posterior lumbar interbody fusion.[31] These findings of rhBMP-2 harms and questionable benefits are consistent with the prior review that precipitated the YODA project,[32] underscoring the need for systematic and complete reporting of benefits and harms, skepticism about the clinical benefits of new technology until evidence emerges from scientifically sound studies, and for patients to be fully engaged in decisions about their care.

Management of Neck Pain

Non-surgical Care: Treatments, Outcomes, and Controversies

There are multiple non-surgical treatments for NP and whiplash-associated disorders (WAD). Non-surgical therapeutic interventions include among others education and advice, exercises, manual therapies (eg, mobilization, manipulation, massage), physical modalities (eg, transcutaneous electrical nerve stimulation [TENS], ultrasound, diathermy, low-level laser therapy), collars, acupuncture, medication (eg, analgesics, steroids, NSAIDs), or combinations of treatments (eg, mobilization and exercises). In a best evidence synthesis (1980-2006), it was found that educational video exercises and mobilization were more effective than usual care or physical modalities for WAD.[33] For other NP, supervised exercises and manual interventions, low-level laser therapy and perhaps acupuncture were more effective than other treatments, or no treatment, although none of the treatments were judged as superior in the short or long run.[33] In particular, evidence of long-term effects was lacking for all non-invasive (non-surgical) interventions and the optimal amount of non-surgical treatment needed was also unclear, as no dose-response (ie, higher frequency of treatments results in better outcomes) or dose-duration (ie, longer duration of care results in better outcomes) relationships were detected. In addition, insufficient data on effective non-surgical interventions for acute non-traumatic neck disorders and Grade III neck pain (disorders with radiation and neurologic signs) precluded any meaningful clinical inferences.[33]

Surgical Care: Treatments, Outcomes, and Controversies

Surgical care is often recommended for patients with NP; however, surgery is costly and exposes patients to inherent serious risks and perioperative pain, morbidity, and complications. When care focuses on conditions of specific pathologic etiology, surgical treatments could be the only solution (eg, after some acute injuries, neoplasms or spinal infections). Most NP cases, however, are not accompanied by aggressive pathology and are not in need of immediate surgical intervention. Nevertheless, surgical care is often performed for NP without serious underlying pathology and its effectiveness is not currently well understood.[34]

In a best evidence synthesis of the limited available literature on surgical interventions for neck pain without serious underlying pathologic conditions, support was lacking for open or percutaneous surgeries in patients with only NP without radicular symptoms or evident serious pathology, while surgical treatment and injections for cervical radicular symptomatology might be appropriate for patients with severe impairments.[34]

Discussion

Enhancing Prevention, Improving Prognosis, and Reducing harms

Based on current knowledge it is not possible, in most cases, to prevent a first-ever (primary) episode of back or neck pain.[4] Recurrences, however, together with the disability and remaining "costs" to the individual and society could be the target of prevention efforts. Health promoting recommendations for multiple chronic diseases, such as physical activity, smoking cessation or maintaining a healthy weight, might also have protective effects on back pain.[4] Having said that, we know that we have a lot of work to do, as for example currently none of

the interventions are backed up with strong evidence for their effectiveness in preventing recurrences of back pain, with the possible exception of physical exercise in reducing the number and duration of recurrences.[4]

It is challenging to study and treat spine disorders due to variability in clinical presentation, in case definitions used, in course or prognosis, as well as in treatments and limited success in their effectiveness.[10] However, new models of care, borrowed from other research areas, are being introduced in spine research. For example, stratified care aims at optimizing effects, minimizing harm and increasing healthcare efficiency.[10] In such a model, it is suggested to use: (a) prognostic stratification to guide treatment (eg, reassure or offer minimal treatment to patients with low risk for poor outcomes, but more extensive treatment to those with a high risk for poor outcomes); (b) patient characteristics to guide treatment; and (c) targeted treatment to the patients most likely to respond (eg, surgery on evidence-based indications), or less likely to be harmed (eg, opioid avoidance for those at risk of dependency).[10]

Conclusion

There is no reason to expect that efforts will fail in enhancing prevention, improving prognosis, and reducing harms related to LBP and NP. In fact, efforts to reduce the burden of LBP and NP might have the same or similar solutions: (a) harmonize LBP and NP case definitions and recommendations between national and international agencies, associations, organizations, and researchers; (b) conduct randomized trials to determine the effectiveness of popular interventions that lack evidence of effectiveness; (c) research all domains and domain-specific outcomes in the short- and long-term;[23] (d) explore new models of care and conceptual models that go beyond the characterization of acute or chronic episodes of pain, ie, consider pain trajectories and a life-course approach;[4] (e) increase our knowledge of spine disorders in both developed and developing countries; (f) assure that evolving LBP and NP in the developing world receive the appropriate recognition and funding to implement culturally acceptable approaches;[4] and (g) identify safe, effective, and cost effective interventions, as rising spine-related health expenditures have not been accompanied by population health improvement in LBP or NP.

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